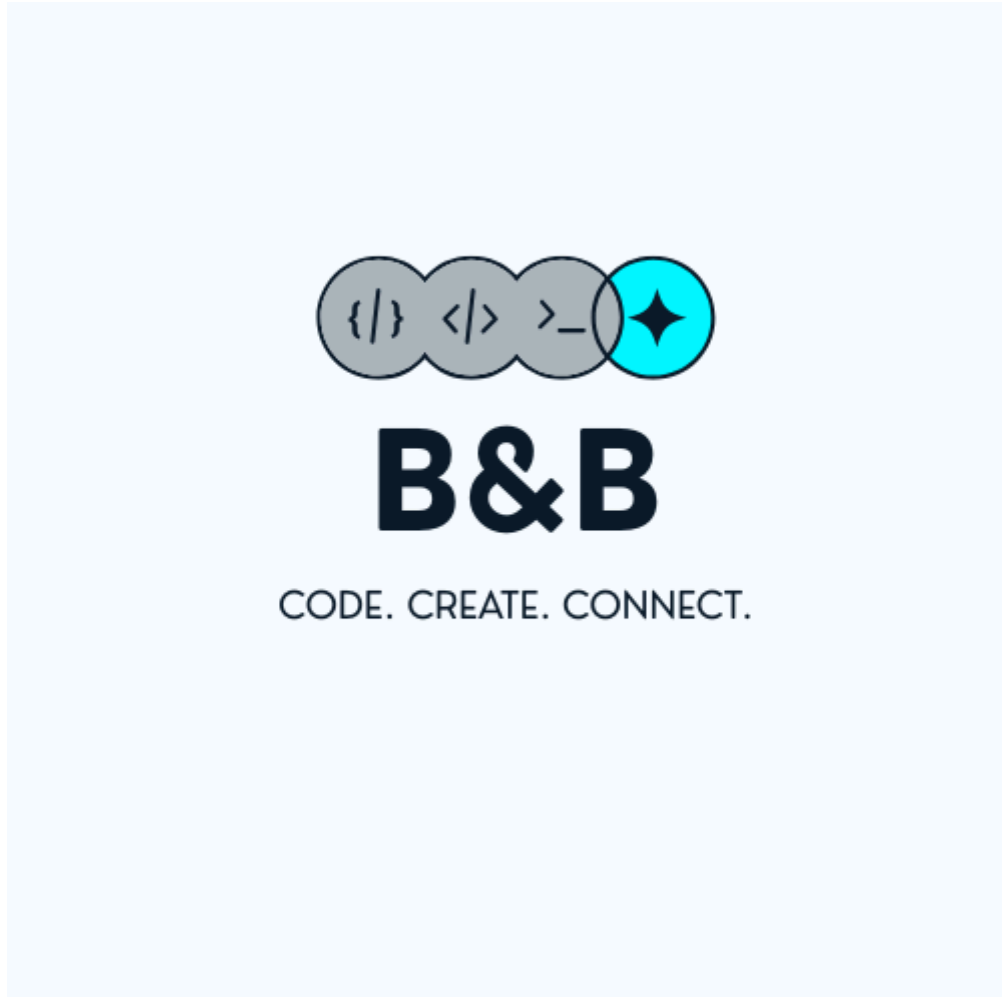


User Evaluation

Team 8: Ben and Bensons



Team Members: Alyssa Skipper, Benjamin Senior, Benson Chow, Chloe Ward, Florian Mengkris, Hannah Thomas, James Ingram, Olivia Spencer

Introduction

Our approach to User Evaluation followed the process of Task Based User Evaluation (TBUE). Users are to complete a series of pre-defined tasks set to model the core functionality of our maze game. All users are given the same set of tasks, such as exploring the maze and interacting with events, to ensure consistency across evaluations. Their responses and any issues they encountered are recorded and compiled to inform future design recommendations.

Recruitment

The target audience outlined by our client defines users as prospective University of York students attending open days as an engaging introduction to the subject. However due to practicality constraints, our project's user evaluation was constrained to current students within our ENG1 cohort. The cohort belongs to a similar age group and demographic which provides a suitable approximation to the intended users, although familiarity with the game may reduce the authenticity of results. To recruit users, our team utilised our scheduled lab time to find students willing to participate in our user evaluation. All potential participants were properly informed on the type and number of tasks involved as well as the expected duration of the session.

Data Collection

In order to complete user evaluations within the allotted time frame we decided to conduct the tests online through Zoom due to the fact that the game was not ready for in-person testing during the semester. Testing remotely allowed a version of the game with all implemented events to be tested, allowing feedback to be gathered on all prototype core features. Initially, after determining the most crucial features, our team compiled a series of tasks to be completed within the interviews. All participants were then made to fill in a consent form on google forms in accordance with ethical guidelines. As tasks were completed the think-aloud protocol was initiated - users were prompted to communicate their thought process while running through the game. As participants shared their thoughts, a scribe recorded key observations and comments. If an issue arose, users were then asked to rate the severity of the problem. Subsequent to completing the test, participants were sent a google form to complete. This allowed them to provide feedback in their own time and helped reduce any social pressure provided during the interview.

Procedure

Each user evaluation session followed a consistent structure. Initially, participants were welcomed and briefed of the study purpose. Participants were informed that the evaluation's purpose focused on the usability of the system - not their own ability. They were then guided through a series of predefined tasks while utilising the think-aloud protocol. All their thoughts, errors and comments were noted down by a scribe. This feedback was then consolidated into a usability table, each issue experienced was noted, along with the user(s) affected. The *severity rating [1]* from the user was then recorded to help prioritise the most important issues in a typical user experience . Severity can be defined on a scale from 1 - 4

1. **Cosmetic Problem** - does not affect functionality
2. **Minor Problem** - affects usability but does not prevent progression
3. **Major Problem** - significantly affects usability and some progression
4. **Catastrophic Problem** - prevents completion

The mean severity rating was taken (rounded) and recorded within the Usability Table. Feedback was relayed to the implementation team and most usability issues were resolved.

Usability Table

ID	Description	Frequency (users)	Severity Rating (1-4)
1	User(s) sometimes struggled initially with understanding how to progress. Some users incorrectly went left into the maze instead of entering langwith. This leads to confusion and delayed progression.	4/8	3
2	User(s) discovered a part of the path that was supposed to be accessible but was blocked by an invisible barrier.	1/8	2
3	User(s) initially questioned if the game had audio. The in-game audio default volume was too low. This can ,however, be resolved by entering the menu and turning up the volume.	2/8	1
4	When entering Ron Cooke, a user discovered there was no upper boundary. This created minor confusion in the playable area.	1/8	1
5	Worms occasionally spawned in unintended locations. This included just before the bus stop, which blocked the path and prevented players from completion.	2/8	4
6	Instructions with night goggles paired with torch instructions exceeded the screen bounds, causing part of it to be cut off. This affected users that found the hidden events.	5/8	2
7	The occurrence of some events were unclear. Users expressed the need for more on screen explanations to prevent confusion.	6/8	2
8	User(s) found the use of some of the items to be unintuitive. In particular,the function of the ghost potion was unclear, and led to confusion when encountered.	4/8	2
9	Several users experienced difficulty when entering buildings due to unclear entrances at the Ron Cook and Piazza. This problem was sometimes worsened after acquiring the ghost potion, which let users run over the buildings before actually entering them.	4/8	3
10	User(s) are unable to access the achievements page before completing the game.	8/8	2
11	User(s) found that if they collided with a worm after the goose had disappeared, the worm was not eaten. This resulted in the player becoming permanently stuck, forcing the player to restart. Users found this confusing and frustrating.	3/8	4
12	User(s) sometimes found it difficult to pick up items despite being on top of them.	4/8	2

[1] C. Vasiliou, "Engineering 1: Software & Systems Engineering, Week 11 – User Evaluation," Lecture, Dept. of Computer Science, University of York, 2025.